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Hyperspectral Band Selection Using Metaheuristics for the Detection of *Aspergillus flavus* in Figs with Machine Learning

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ABSTRACT

The contamination of figs by *Aspergillus flavus* is a critical problem for food safety and the agricultural economy due to the production of carcinogenic aflatoxins. Hyperspectral imaging offers a non-invasive solution for detection, but its high dimensionality represents a computational challenge. This work proposes a hybrid approach to optimize detection by selecting the most informative spectral bands. Two metaheuristics Genetic Algorithms and Particle Swarm Optimization are used to identify an optimal subsets of three, five and six spectral bands. The fitness of each subset is evaluated using the classification F1-score of a Machine Learning classification algorithm Support Vector Machine. This method is expected to drastically reduce data dimensionality, lowering computational cost without sacrificing accuracy and laying the groundwork for real-time inspection systems.

1 | Introduction

Food safety and economic sustainability in agriculture face a constant threat due to contamination by pathogenic fungi. Among them, *Aspergillus flavus* stands out for its dangerousness, as it is the main producer of aflatoxins, secondary metabolites classified as Group 1 human carcinogens, closely linked to the development of hepatocellular carcinoma [2]. The detection of this pathogen is critical, especially in high-value crops such as the fig (Ficus carica), where regions like Extremadura (Spain) lead national and European production [2]. Traditionally, the identification of these mycotoxins has relied on analytical chemical methods (such as HPLC or ELISA), which, although precise, are destructive, expensive, and time-consuming, preventing the inspection of 100% of the production [1, 12]. To overcome these limitations, the industry has turned towards non-invasive assessment methods. In this context, Hyperspectral Imaging (HSI) has emerged as a revolutionary tool. Unlike conventional computer vision, HSI technology captures detailed spectral information across hundreds of contiguous bands for each pixel of an image, allowing the detection of subtle physicochemical changes in plant tissues before symptoms become visible to the human eye [17, 1]. However, the large amount of data generated by these images presents a significant challenge for their efficient processing.

The use of HSI for the detection of fungal diseases has been widely explored in the last decade. Previous research has demonstrated the efficacy of this technology in detecting aflatoxins and fungi in maize, peanuts, and other nuts [12, 2].

With respect to figs, recent studies have applied deep learning techniques, such as Convolutional Neural Networks (CNN) and Fully Connected Neural Networks (FCNN), to detect the infection of *Aspergillus flavus* [2]. The results presented high accurary rates for classifying infected pixels. This research estudy set an important precedent by comparing deep neural network architectures on the same type of samples, and demonstrating the viability of a pixel-level analysis to accurately predict the presence of *Aspergillus flavus*.

However, many of these approaches utilize the full spectrum, which requires expensive hardware and high computational power. Current literature suggests that spectral band selection is a crucial step for practical implementation [14]. Statistical methods such as Principal Component Analysis (PCA) have traditionally been used to reduce data [1], but techniques based on evolutionary algorithms (such as GA and PSO) are gaining ground due to their ability to find global optimal solutions in complex search spaces [14].

The main objective of this research work is to optimize the detection of the fungus *Aspergillus flavus* in fresh figs using an evolutionary approach. The development of the engineering features strategy involved the implementation of a genetic algorithm as an intelligent spectral band selection. Support Vector Machine (SVM) was applied to assess the fitness of the individuals.

Based on the advances obtained, this work concludes that the fusion of mean reflectance and standard deviation data allows achieving performance metrics (F1-Score) exceeding 83%.

The results demonstrated that the GA identified an optimal set of five spectral bands, predominantly in the visible spectrum, with bands near the ultraviolet and near-infrared regions. These findings validate that it is possible to drastically reduce the complexity of hyperspectral data without compromising detection capability, offering a promising path for the agricultural industry.

2 | Materials and Methods

The methodology employed in this work is structured into three sequential stages, as illustrated in Figure 1. First, the acquisition of hyperspectral images of healthy and infected figs was carried out. The second stage consisted of image preprocessing, which included radiometric correction, region of interest (ROI) segmentation, and the extraction of representative statistical metrics (mean and standard deviation) per sample. Subsequently, an optimization strategy based on metaheuristics was implemented, evaluated using a Support Vector Machine (SVM), with the objective of reducing dimensionality and selecting the optimal spectral bands for the detection of *Aspergillus flavus*.

2.1 | Sample Preparation and Data Acquisition

For data acquisition, the same hyperspectral images as in the work of [2] were used.

These images were generated using fresh figs (Ficus carica L.) of the *Calabacita* variety, which were harvested at the "Finca La Orden-Valdesequera" belonging to the Scientific and Technological Research Center of Extremadura (CICYTEX), located in Guadajira, Badajoz, Spain. All the figs were harvested at the same time.

2.1.1 | Inoculation process

The inoculation process was carried out under controlled conditions by agriculture engineers belonging to CICYTEX. The figs were arranged into four trays, each containing 36–40 fruits. One tray served as the healthy control group, while the figs in the remaining three trays underwent the inoculation process. As a first step, hyperspectral images for healthy controls were taken.

The inoculation involved submerging the lower portion of each fig in a solution containing a strain of *Aspergillus flavus* for 4–5 seconds. This process ensures that the distribution of the fungus on the fig surface is comparable to natural contamination. Particularly, for this study, one strains from the collection of the CAMIALI group from the University of Extremadura, Spain (*Aspergillus flavus* HG 144M) was used.

The protocol set during the inoculation process was designed to apply a solution containing three different concentrations of the fungi to the figs in the three specific trays. These concentrations were 10^3 UFC/ml, 10^5 UFC/ml, and 10^7 UFC/ml, identifying classes 1, 2 and 3, respectively. Quantification of *Aspergillus flavus* was performed in CFU/mL to estimate the viable fungal load, considering only those units capable of proliferating on the cultivation medium. This approach is particularly relevant for assessing the potential for colonization.

Hyperspectral images were acquired every 24 hours for five consecutive days following inoculation. Between the acquisition sessions, samples were kept in a controlled-environment incubation chamber. The chamber was set at 25 °C with a relative humidity of $80-90\%$, providing environmental conditions optimized for fungal growth and development. Once data acquisition was finished the first subsequent downstream analysis started [2].

Image acquisition was performed using a line-scan (pushbroom) hyperspectral camera, particularly the SPECIM model FX10. The sensor of this camera captures information in the visible and near-infrared (VNIR) spectral range. This device features a spectral range from 400 nm to 1000 nm, and a spectral resolution of 448 bands with a spectral sampling interval of approximately 1.339 nm. Its spatial resolution is 1024 pixels in spatial width per scan line [15].

In this work, not all images for all classes were used. The focus was on discriminating between healthy controls and samples inoculated with the high concentrations of the fungi, identified as class 3 ($10^7 UFC/ml$), which constituted the working dataset. This study addressed the binary classification between infected and non-infected samples. From now on, we will refer to these as classes 0 and 1, respectively.

As a summary, these samples are organized as follows:

- Class 0: Healthy controls with non-inoculated figs, used as a reference for the healthy state of the fig.
- Class 1: Severe infection class. Figs inoculated with an *Aspergillus flavus* concentration of $10^7 CFU/mL$, representing a high fungal load scenario.

2.2 | Data Preprocessing

As previously mentioned, in the acquired images the figs were arranged on trays, each containing approximately 40 samples, and the evaluation needed to be conducted for each independent fig. To process the spectral information of each fruit individually, manual segmentation was performed using the open-source software LabelMe (v5.8.3) [16]. This process resulted in the processing of 18 hyperspectral scenes, yielding a total of 560 individual segmentation masks. Consequently, the final dataset consisted of 560 samples, distributed as 277 samples for the control class (Class 0) and 283 samples for the inoculated class (Class 1) with a spore concentration of 1×10^7 CFU/mL.

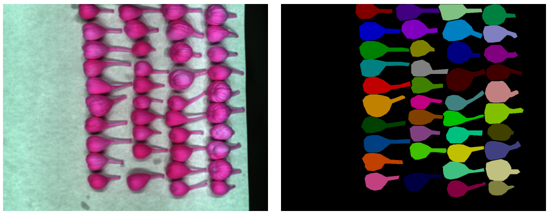

FIGURE 1 | Figs tray before segmentation (**a**) and fig masks after segmentation (**b**).

Data processing and feature extraction were conducted using the Spectral Python (SPy) library. First, the header (.hdr) and binary (.raw) files were read to reconstruct the hyperspectral data cubes. Raw intensity values were then converted into relative reflectance through radiometric calibration, using the white and dark references acquired with each scene, according to equation [8]:

$$R = \frac{I_{raw} - I_{dark}}{I_{white} - I_{dark}} \tag{1}$$

where I_{raw} is the raw image, I_{dark} is the dark reference, and I_{white} is the white reference. Following calibration, spectral features were extracted from the previously segmented ROIs. For each sample (fig), the mean and standard deviation of the reflectance were calculated across all spectral bands. These values were stored in two distinct feature matrices; one representing the average spectral signatures and the other capturing spectral variability, which served as inputs for the subsequent band selection stages.

2.3 | Dimensionality Reduction and Band Selection

Although hyperspectral imagery provides abundant information, it suffers from significant redundancy due to the high correlation between contiguous spectral bands [14]. This characteristic leads to the called *curse of dimensionality* or Hughes phenomenon, where classification accuracy tends to decrease as the number of spectral bands increases, given a limited number of training samples [6]. This high redundancy is shown in the Figure 2 which is a correlation matrix of the spectral bands in the dataset generated of mean reflectance. The prominent structure formed in the diagonal of the matrix indicates groups of correlation coefficients near to 1.0. Furthermore, the high data volume significantly increases the computational burden required for processing. To address these challenges, band selection has proven to be an effective strategy for dimensionality reduction, eliminating redundancy while preserving the most discriminative spectral features [14, 10]. In this study, two metaheuristic approaches are employed for optimal band selection: Genetic Algorithms (GA) and Particle Swarm Optimization (PSO).
Comparing PSO and GA can provide valuable information to the problem at hand. PSO is more effective in exploiting the search space by quickly adjusting solutions near the optimum, but it can get trapped in local optima due to its tendency to rapidly focus on specific areas. In contrast, GA favors global exploration and diversity, avoiding premature convergence and being more robust in multimodal spaces. Comparing both algorithms can help to identify whether the problem will require more exploration, where GA excels, or more exploitation, where PSO is more efficient. Not only will this improve the efficiency and robustness in the search for global optima, but it also can provide valuable insights to guide the next research steps.

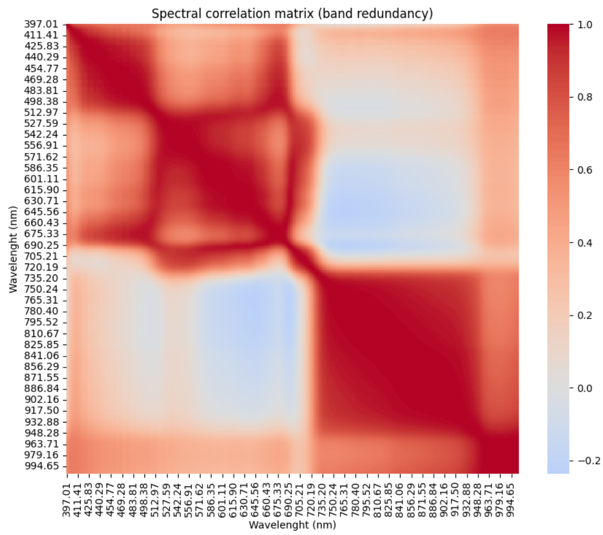

FIGURE 2 | Figs mean values correlation matrix.

2.3.1 | Feature Construction and Optimization Problem.

To thoroughly capture the impact of fungal infection on both spectral response and surface texture, a concatenated feature vector was constructed combining the mean (μ) and standard deviation (σ) of the reflectance values across all the pixels within the ROI for each spectral band. The mean reflectance characterizes the central tendency of the spectral signature [10], while the standard deviation quantifies the spatial variability of the sample surface [5]. Formally, for a hyperspectral image with n bands, the complete pool of available features for the i-th sample is defined as $Xi = [\mu_{i,1}, ... \mu_{i,n}, \sigma_{i,1}, \dots, \sigma_{i,n}]$. However, to maintain spectral consistency, the dimensionality reduction process was formulated as a band selection problem rather than an individual feature selection task. This means that selecting the j-th spectral band implies the simultaneous inclusion of both its corresponding mean (μ_j) and standard deviation (σ_j) into the classifier input vector.
Therefore, the optimization search space consists of n binary variables (where n is the number of bands), but the dimensionality of the resulting feature vector sent to the classifier is $2{\times}k$, where k is the number of selected bands. The objective is to find the binary vector $B = \{b_1, \dots, b_n\}$ that maximizes the F1-Score (2).

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{2}$$

The F1-Score was selected as the fitness function due to its ability to balance precision and recall. It is critical to penalize false negatives (infected figs misclassified as healthy), as these represent a significant health risk. Furthermore, the F1-Score provides a robust performance metric that effectively mitigates the bias potentially introduced by the slight class imbalance in the dataset.

2.3.2 | Genetic Algorithm

The Genetic Algorithm (GA) is a metaheuristic optimization technique bio-inspired by natural evolution and genetic recombination. It encodes candidate solutions (individuals) as vectors and iteratively applies evolutionary operators such as selection, crossover, and mutation, often supplemented by elitism strategies [4, 3]. This implementation used a 448-bit binary string to encode the band subset, where a value of "1" indicates selection, and a "0" value denotes exclusion. The algorithm evolves the population over successive generations to optimize the target fitness function. The specific parameter settings used in this study are detailed in Table 1.

2.3.3 | Particle Swarm Optimization Algorithm

Particle Swarm Optimization (PSO) is a stochastic metaheuristic inspired by the collective social behavior of bird flocks and fish schools. Unlike traditional evolutionary algorithms that rely on recombination or crossover operators, PSO models the population members as particles traversing an n-dimensional search space. By analogy, a swarm is similar to a population and a particle is similar to an individual. Each particle is characterized by its current position (representing a candidate solution for the band selection) and a velocity vector that governs its trajectory. The movement of a particle is determined by updating its velocity based on a weighted sum of three distinct components (inertia, cognitive and social) [4].
To enhance convergence efficiency and mitigate premature convergence to local optima, Time-Varying Acceleration Coefficients (TVAC) were implemented. The inertia component was set at a fixed value. The cognitive coefficient ($C1$) was linearly decremented to encourage exploration during the early stages, while the social coefficient ($C2$) was linearly incremented to promote global exploitation towards the end of the search process. The dynamic adjustment of these parameters follows standard literature recommendations for high-dimensional optimization [13].

$$c_1(t) = (c_{1,f} - c_{1,i})\frac{t}{T_{max}} + C_{1,i} \tag{3}$$

$$c_2(t) = (c_{2,f} - c_{2,i})\frac{t}{T_{max}} + C_{2,i} \tag{4}$$

Where:

- t: Current iteration.
- T_{max}: Maximum number of iterations.
- $C_{1,i}$,$C_{2,i}$: Coefficient initial values.
- $C_{1,f}$,$C_{2,f}$: Coefficient final values.

The specific parameter settings used in this study are detailed in Table 2.

2.4 | Classification Model Implementation

A Support Vector Machine (SVM) was implemented for the supervised classification task, selected for its proven robustness in high-dimensional spaces and effectiveness with limited datasets [10, 9]. Given the non-linear nature of the spectral boundaries between healthy and infected samples, the Radial Basis Function (RBF) kernel was employed.

2.4.1 | Computational Strategy

To mitigate the high computational overhead inherent to the iterative metaheuristic process (GA and PSO), a two stage evaluation strategy was implemented:

1. Fast evaluator: A lightweight SVM configuration used as the wrapper within the fitness function. Its primary goal is to rapidly estimate how well perform the selected bands to guide the search algorithm, prioritizing the execution speed over hyperparameter tuning.
2. Exhaustive evaluator: A rigorous final model, fine tuned via exhaustive hyperparameter search.

Before training, data normalization, using StandardScaler, was applied to ensure that both feature types (mean reflectance and standard deviation) contribute equally to the distance calculation. Finally, to ensure an unbiased evaluation, the winning model is evaluated on a test set that remained unseen during the optimization phase. The configuration parameters for both implementations are detailed in Table 3.

3 | Results

3.1 | Spectral Characterization of Samples

Figure 3a displays the average spectral signatures relative to the healthy and infected fig samples across the full acquired spectral range (400–1000 nm). The shaded regions represent the standard deviation, illustrating the spectral variability within each class. As observed, the spectral patterns of both classes are visually similar, particularly in the visible region (400–700 nm). However, slight differences in reflectance intensity become more apparent in the Near Infrared (NIR) region (700–1000 nm), likely due to the structural degradation of the fig tissue caused by fungal colonization. This high spectral overlap confirms the complexity of the classification task. Figure 3b presents the standard deviation per wavelength. It is observed that infected samples have a major variability compared to the healthy ones. This visual evidence leads to support the premise that the fungal infection affects the textural heterogeneity on the fig skin.

3.2 | Impact of Feature Construction

To validate the contribution of the standard deviation to the fungal classification, the band selection process was executed under two distinct feature configurations. Initially, the optimization was performed using only the mean reflectance per band, followed by a second scenario utilizing the combined mean and standard deviation vector. The optimization was conducted for target subset sizes of $k \in \{3, 5, 6\}$ over 30 independent runs. Table 4

TABLE 1 | Genetic Algorithm parameters.

Generations	Population size	Selection type	Crossover	Mutation rate
350	250	Tournament of 2	Single Point	2.5

TABLE 2 | PSO parameters.

Iterations	Particles	C1	C2	Inertia	Mutation rate
650	200	0.4 to 2.5	2.5 to 0.4	0.7	0.2

TABLE 3 | SVM Parameters.

Implementation	Kernel	Regularization (C)	Gamma (γ)	Validation method
Fast evaluator	RBF	Fixed 1.0	Fixed scale	k-fold CV ($k = 3$)
Exhaustive evaluator	RBF	Grid search 0.1 to 1000	Grid search 0.001 to 1	k-fold CV ($k = 5$)

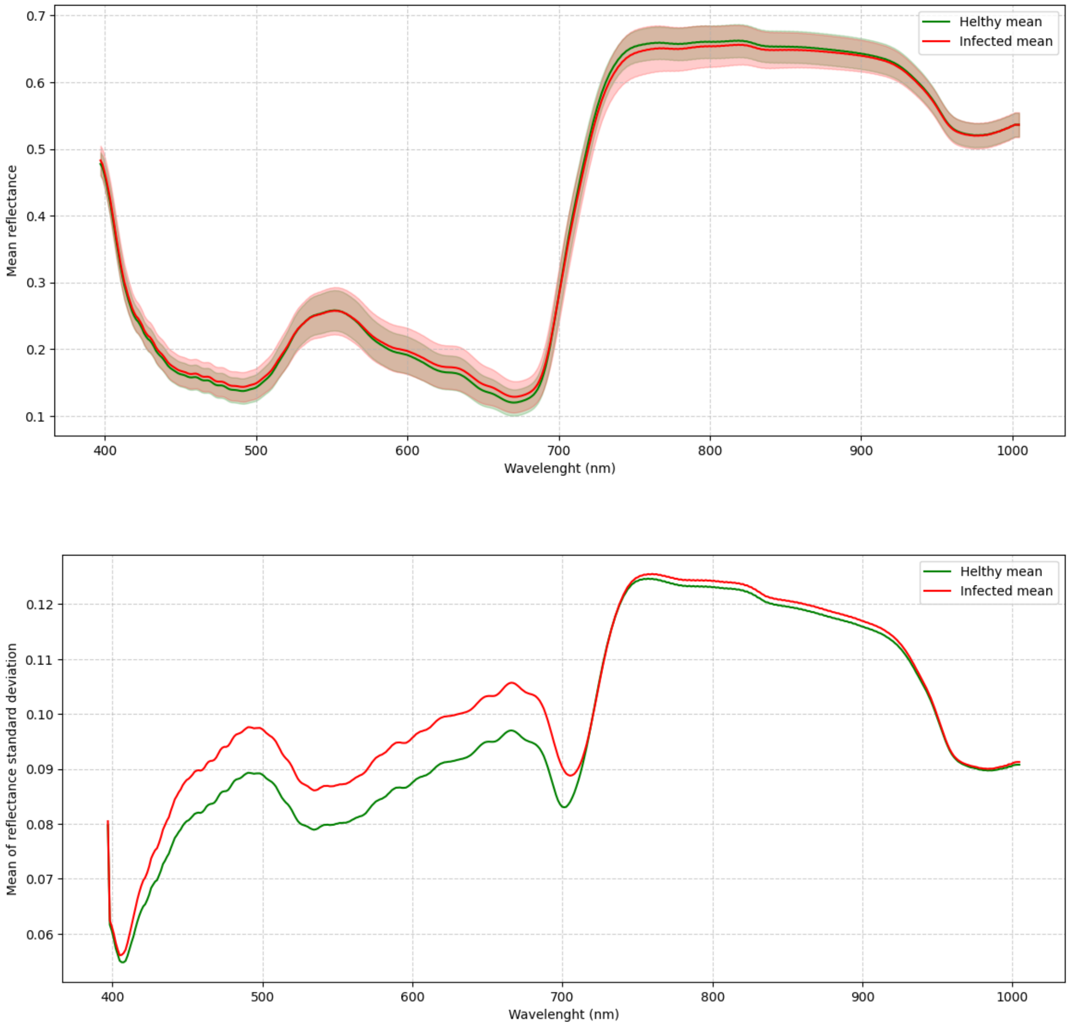

FIGURE 3 | Spectral characterization of the fig samples. (**a**) Average reflectance signatures. (**b**) Standard deviation signatures.

compares the average F1-Score obtained by the GA across both configurations.

In addition to the previous performance metrics, the evolutionary behavior of the optimization process was analyzed. Figure 4 illustrates the average convergence history over 30 independent

TABLE 4 | GA performance comparison (average F1-Score) between feature configurations.

Individual size	Mean reflectance only (μ)	Combined features ($\mu + \sigma$)	Improvement (%)
3 Bands	0.663	0.755	+9.2%
5 Bands	0.698	0.77	+7.2%
6 Bands	0.721	0.773	+5.2%

runs for both feature configurations. The solid lines represent the mean best fitness value at each generation, while the shaded regions indicate the standard deviation across runs. It can be observed that the Combined Features ($\mu + \sigma$) strategy (represented by the blue line) not only starts with a higher initial fitness but also maintains a superior trajectory throughout the generations compared to the Mean Reflectance Only approach. This convergence gap visually confirms that the additional textural information provides a more robust search space, allowing the proposed algorithm to find higher quality solutions more efficiently.

3.3 | Comparative Analysis of Metaheuristic Algorithms

After the validation of the feature construction, the study focused on comparing the optimization capabilities of the GA versus PSO. To ensure statistical robustness, both algorithms were executed 30 times for each target subset size ($k \in \{3, 5, 6\}$) using the combined feature vector.

The figure 5 presents the boxplots summarizing the distribution of the F1-Scores obtained. The analysis reveals that the GA has a marginal difference with PSO in the context of hyperspectral band selection. Specifically, GA exhibited a higher median F1-Score when $k = \{5, 6\}$ subset sizes, particularly in the most scenario (k=6), where it achieved a median value of 0.773. However, PSO had a slight advantage in the 3 bands scenario. Furthermore, the interquartile ranges for GA were consistently narrower compared to PSO, indicating superior algorithmic stability and reproducibility.

Notably, the absolute maximum F1-Score of the entire study (0.836) was achieved by PSO using the 5-band subset with the ($\mu+\sigma$) configuration. However, it is noteworthy that the configuration using only the mean (μ) with 6 bands yielded a nearly identical score of 0.832. This suggests that the inclusion of the standard deviation stabilizes the search process and improves classification performance when the feature subset is small (e.g., 3 bands); however, this advantage diminishes as more bands are added. Figure 6 illustrates this phenomenon, showing how the variance in the solution space increases for the 6-band configuration that relies solely on the mean.

4 | Discussion

4.1 | Physical interpretation

The spatial distribution of the selected bands in the best individuals was analyzed against the spectral signatures of the samples. Figure 6 illustrates the mean reflectance spectra for healthy (green) and infected (red) figs, with the vertical blue dashed lines representing the bands of the best individuals selected by the GA across the 30 execution runs for $K \in \{3, 5, 6\}$.

The visualization reveals a not uniform distribution of selected bands; the bands cluster in three different spectral regions that correspond to known physiological changes caused by plant diseases.

1. 400 – 450 nm: A cluster of bands is observed in the lower visible spectrum. This region is associated with the absorption of pigments such as carotenoids and anthocyanins [7].
2. 700 – 750 nm: This accumulation of selected bands around the red edge (between visible red and near infra red) is sensitive to stress induced changes in cellular structure [7].
3. 950 - 1000 nm: The selected bands in the end of the selected spectrum coincides with second overtone of the O-H bond stretch, indicating possible signs of dehydration [11].

4.2 | Conclusions and Future Work

This study presented a non invasive methodology for the detection of *Aspergillus flavus* in figs using hyperspectral imaging coupled with metaheuristic band selection. By constructing a statistical feature vector based on the spectral mean (μ) and standard deviation (σ), the proposed approach successfully reduced the high dimensional hyperspectral data into a compact subset of informative variables.

4.2.1 | Key findings

1. Algorithmic Performance: In the majority of cases, the GA exhibited slightly better average performance compared to PSO. However, PSO identified the global maximum, the absolute best solution. This indicates that both metaheuristics possess comparable optimization capabilities for this spectral selection task.
2. Feature relevance: The results highlighted that, on average, the inclusion of the standard deviation ($\mu + \sigma$) significantly improved performance in lower dimensional subsets ($k \in \{3, 5\}$). However, as the number of selected bands increases, this performance gain becomes marginal, suggesting that the added complexity of the standard deviation is less critical when more spectral information is available.
3. Physical Consistency: The spectral bands selected correspond with relevant biological regions; they are aligned with the visible pigmentation changes and NIR water content regions.

4.2.2 | Future work

Having the current results in mind several paths for future work have been identified to further enhance the system performance:

- Multi-Objective Optimization: Instead of constrain the search to a fixed or arbitrary number of bands, as done

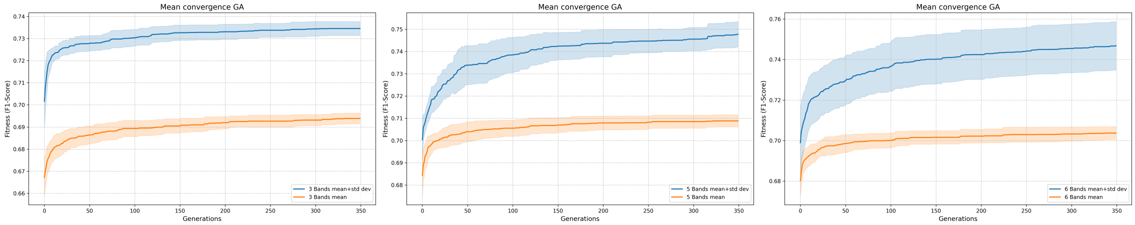

FIGURE 4 | Genetic Algorithm convergence profiles: (**a**) 3-band; (**b**) 5-band; and (**c**) 6-band optimization scenarios.

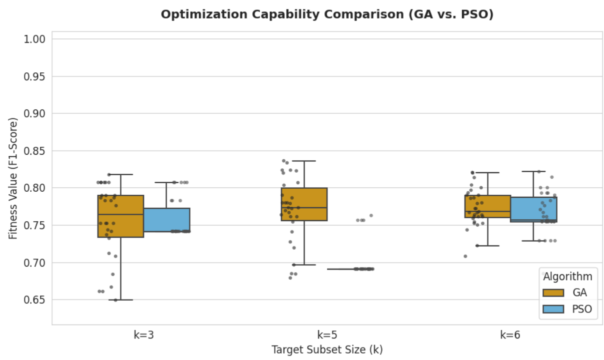

FIGURE 5 | Boxplot comparison GA vs. PSO with ($\mu + \sigma$) features.

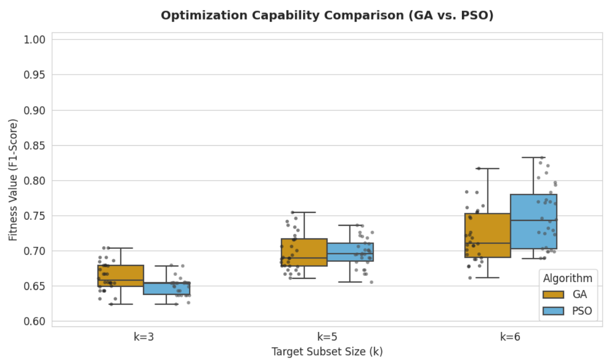

FIGURE 6 | Boxplot comparison GA vs. PSO only with (μ) feature.

in the present work, future iterations might employ multi-objective optimization algorithms. The goal is to maximize the F1-Score and minimize the number of bands, thereby finding the optimal Pareto front that balances classification accuracy with system simplicity.

- Advanced Classifiers: While the SVM classifier used demonstrated solid performance, we aim to explore more other machine learning algorithms such as Random Forest and XGBoost, which may offer better handling of the non linear interactions of the spectral data.
- Co-evolutionary Strategies: To improve the feature selection process itself, we propose exploring co-evolutionary techniques. These methods could allow the population of solutions (bands) and the population of strategies (classifier parameters) to evolve together, potentially discovering more effective spectral combinations that a standard evolutionary approach might miss.

Author Contributions

Israel Calderón Aguilar: Conceptualization, Methodology, Software, Formal Analysis, Visualization, Writing – Original Draft. **Juan Villegas Cortez:** Supervision, Validation, Writing – Review & Editing. **Josefa Díaz Álvarez:** Supervision, Validation, Writing – Review & Editing. **Román Anselmo Mora Gutiérrez:** Supervision, Validation. **Francisco Fernandez De Vega:** Supervision, Validation. **Francisco Chávez de la O:** Supervision, Validation. **Arturo Zúñiga López:** Supervision, Validation.

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Conflict of Interest

The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

GA	Genetic Algorithm
HSI	Hyperspectral Imaging
PSO	Particle Swarm Optimization
SVM	Support Vector Machine

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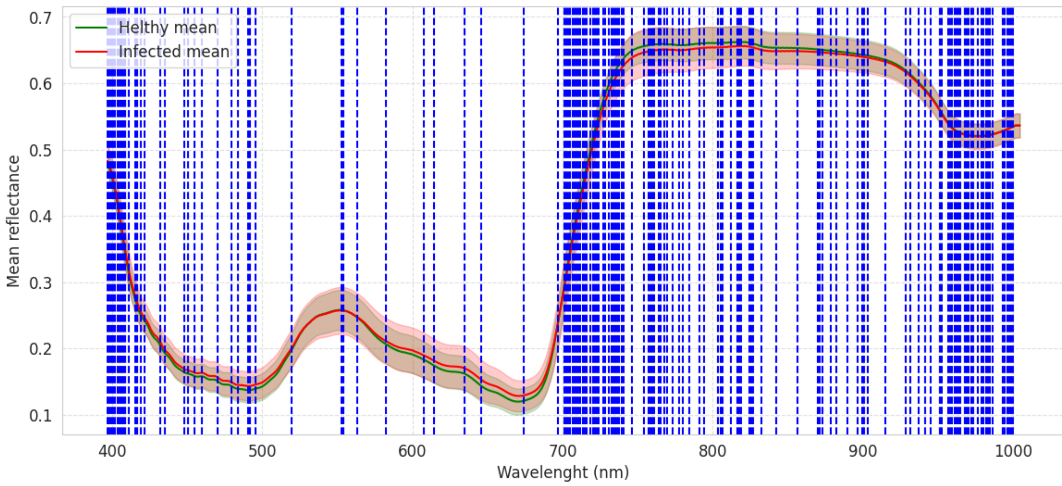

FIGURE 7 | Mean reflectance spectral signature with vertical lines indicating the selected bands by the best individuals of the GA.

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